

Feasibility Report: ForgetIt or Travel Bot

1. Introduction

This report assesses the feasibility of developing “Forget it” or “Travel Bot”, a multi-platform application designed to prevent users from forgetting their personal belongings during travel, short visits or daily routines. The core concept involves creating a dynamic checklist that provides reminders, with a future vision for IoT/RFID integration. While the proposed future feature involving RFID has significant challenges, the initial software-based version of the project is recommended for implementation. Recommendations for ongoing strategy include a phased development approach, starting with a MVP to validate the core functionality with target users.

2. Background

Leaving items behind in locations like trains, hotels, or meeting rooms is a common problem and frustrating experience. This occurs due to the pace we operate when we are distracted, changes in routine happen or simple human error, leading often to significant inconvenience or financial loss that builds up. While generic checklist applications exist, they are not designed for the specific context of “last minute” reminders when leaving a location. This project is conceived to address this gap head on. It aims to move beyond static packing lists by providing a proactive, context aware system that ensures users leave with everything they arrived with, transforming a reactive problem into a managed process.

3. Outline of Project

The primary objective of the project is to develop a user-friendly application that reduces the incidence of forgotten items.

Key Objectives include:

- To allow users to create personalized profiles for trips or visits (e.g., “Train to London”, “Weekend visit”).
- To enable users to build dynamic checklists of items for each profile.
- To implement an intelligent notification system that reminds users to check their list as their departure time approaches.

- To provide a simple, tactile interface for a final check-off process on the day of the event.

Envisaged results and deliverables:

- A functional mobile application (for iOS and Android) built with a cross-platform framework.
 - A secure local database to store user data.
 - A working notification system based on time triggers.
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4. Methodology/Method of Analysis

Analysis of tools, techniques and methodologies:

- Technical Analysis: The proposed technologies (.NET Core, C#, SQLite) were evaluated for their capability to meet project objectives, their compatibility with the developer's skillset, and their support for required features like local data storage and notifications.
 - Market Analysis: A review of existing solutions in the application was conducted to identify competitors (e.g., generic checklist apps like Todoist, and packing apps like PackPoint). This analysis focused on identifying gaps in the market and opportunities for differentiation.
 - Operational Analysis: The resource requirements, including development time, skills and costs, were assessed to determine if the project is manageable for a single developer or a small team.
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5. Overview of Alternatives

Various alternative approaches and solutions were considered:

- Alternative 1: Develop a generic checklist application.
 - Advantages: Simpler scope, faster time to market.
 - Drawbacks: Generic application would not address the core problem effectively. It would lack the context-aware reminders (e.g., "Overhead compartment", "Under table" alerts) that define this project's unique value, making it just another competitor in a crowded market.
- Alternative 2: Develop a native application for each platform.
 - Advantages: Better performance and access to the latest platform specific features

- Drawbacks: Requires maintaining two separate codebases, doubling development time and complexity, which is inefficient for a single developer.
- Alternative 3: Focus on RFID/IoT integration from the start.
 - Advantages: Highly innovative and could provide a definitive solution to the problem.
 - Drawbacks: Technically complex, costly and introduces hardware dependency. This approach could derail the entire project as a result of unforeseen technical challenges.

Evaluation: The alternative of using a cross-platform framework like .Net MAUI presents the most balanced path, leveraging existing skills while efficiently targeting multiple platforms. Deferring the RFID component mitigates the highest risks.

6. Conclusion

The analysis leads to the following findings:

- Technical Feasibility: High for the core checklist application. Modern development tools make this project well within the capabilities of a developer proficient in C#.
- Market Feasibility: Moderate. A clear market gap exists for a specialized application, but success is contingent on creating significantly better user experience than generic alternatives.
- Operational Feasibility: High. The project requires minimal financial investment and aligns perfectly with the developer's available skills and resources.

Based on the analysis conducted, the ForgetIt/Travel Bot project, in its initial software-based form is deemed feasible.

7. Recommendation

It is the recommendation of this report that ForgetIt/Travel Bot should proceed with a phased implementation strategy.

- Phase 1: Immediately begin development of a software only MVP using .NET MAUI and C#. The focus should be on validating the core checklist and reminder functionality.
- User Validation: Concurrently, engage with potential users to gather feedback on the MVP and refine feature priorities. This is critical for market success.
- Phase 2: Upon successful validation of the MVP, explore enhancements such as geofencing and pre-built templates.
- Phase 3: The RFID/IoT concept should be treated as a separate, long-term research and development project, not as a core dependency for the initial application launch.